

Honeywell

Total Plant Solution (TPS) System

**Time Link/GPS Antenna
Troubleshooting
Guide**

TP80000

R100

2/03

TotalPlant

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Honeywell International
Industry Solutions
16404 North Black Canyon Highway
Phoenix, AZ 85053
1-800 343-0228

About This Document

References

The following list identifies all documents that may be sources of reference for material discussed in this publication.

Document Title	Doc ID
GPS Antenna User's Guide	TP22000A
Interpret System Clock Functions	L61524 LCN

Contacts

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Symbol Definitions

The following table lists those symbols used in this document to denote certain conditions.

Symbol	Definition
	ATTENTION: Identifies information that requires special consideration.
	TIP: Identifies advice or hints for the user, often in terms of performing a task.
	REFERENCE -EXTERNAL: Identifies an additional source of information outside of the bookset.
	REFERENCE - INTERNAL: Identifies an additional source of information within the bookset.

Symbol	Definition
CAUTION	Indicates a situation which, if not avoided, may result in equipment or work (data) on the system being damaged or lost, or may result in the inability to properly operate the process.
	CAUTION: Indicates a potentially hazardous situation which, if not avoided, may result in minor or moderate injury. It may also be used to alert against unsafe practices. CAUTION symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.
	WARNING: Indicates a potentially hazardous situation which, if not avoided, could result in serious injury or death. WARNING symbol on the equipment refers the user to the product manual for additional information. The symbol appears next to required information in the manual.
	WARNING, Risk of electrical shock: Potential shock hazard where HAZARDOUS LIVE voltages greater than 30 Vrms, 42.4 Vpeak, or 60 VDC may be accessible.
	ESD HAZARD: Danger of an electro-static discharge to which equipment may be sensitive. Observe precautions for handling electrostatic sensitive devices
	Protective Earth (PE) terminal: Provided for connection of the protective earth (green or green/yellow) supply system conductor.
	Functional earth terminal: Used for non-safety purposes such as noise immunity improvement. NOTE: This connection shall be bonded to Protective Earth at the source of supply in accordance with national local electrical code requirements.

About This Document

Symbol Definitions

Symbol	Definition
	Earth Ground. Functional earth connection. NOTE: This connection shall be bonded to Protective Earth at the source of supply in accordance with national and local electrical code requirements.
	Chassis Ground: Identifies a connection to the chassis or frame of the equipment shall be bonded to Protective Earth at the source of supply in accordance with national and local electrical code requirements.

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3. GPS / Time Link Components

3.7 Overview

Introduction

An overview and detailed description of the proper techniques for installation of the GPS antenna is provided in the *GPS Antenna User's Guide*. This document assumes that the user is familiar with the contents of the *GPS Antenna User's Guide* and is looking for some additional help in diagnosing problems with the GPS or Time Link cable components.

1. GPS / Time Link Components

1.0. Overview

Illustration

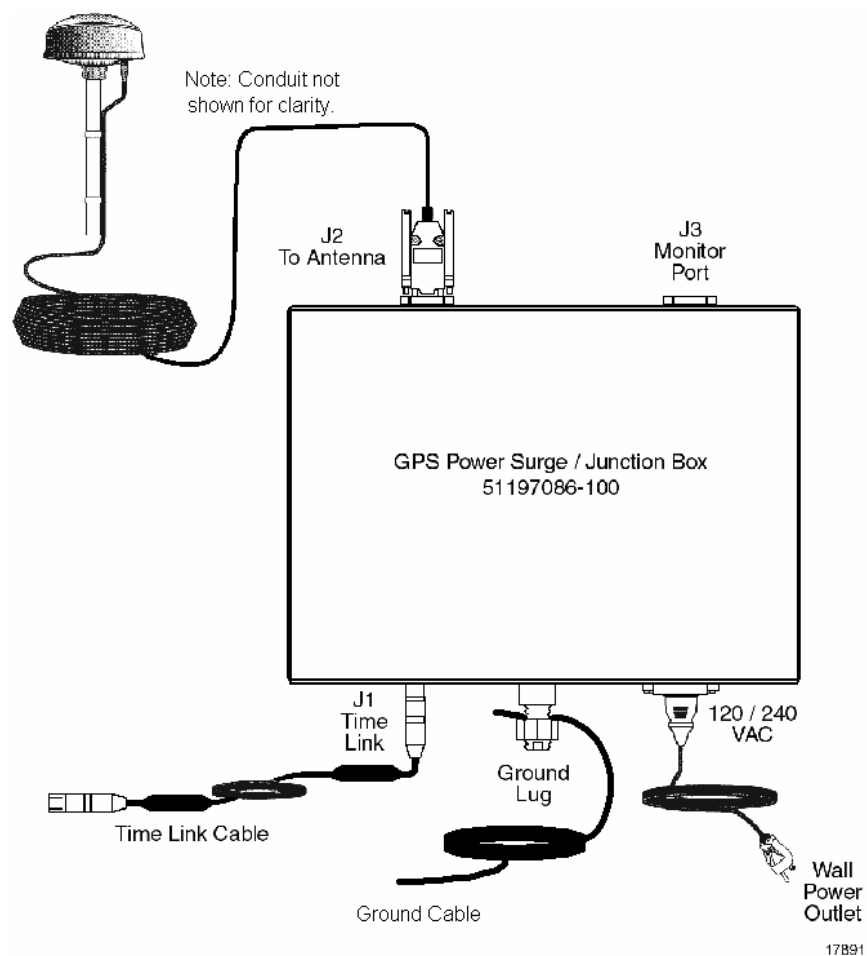


Figure 1: GPS Antenna – Power Junction Box

The Time Link distribution cable consists of CAT5 STP cables, T-connectors and terminators.

The GPS antenna option consists of the GPS antenna, power surge junction box, and cable between the antenna and the junction box.

2.0 Time Link Cables

Description

Time Link cables bring the UTC signal from the junction box to the TPN Bridge or GUS/APP nodes. They are ideal for the TPN Bridge system that requires a low segment-to-segment clock skew. The cable is uniquely identified by a ferrite bead inside adhesive shrink tubing at both ends.

The Time Link cable is available two ways: in standard lengths of 0.5, 2, 5, 10, 20, 50, 75, and 99 meters or custom specified lengths. The Time Link cable is color-coded with Black identification boots (Tab 4XX where XX equals cable length) around each cable end.

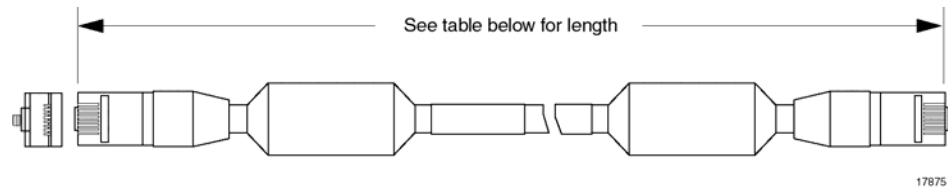


Figure 2 Time Link Cable

Standard Cables Lengths

Standard length cables are available by factory order.



ATTENTION

The sum of all the Time Link cables (junction box to nodes) and the GPS cable (antenna to junction box) lengths is limited to 1000 feet.

1. GPS / Time Link Components

2.0. Time Link Cables

Table 1 Standard Time Link Cable Lengths

Standard Cable Lengths for 100-Base-T Cable w/Black Boots

51305562-4XX Cable Tab	Standard Lengths
400	0.5 meters
402	2 meters
405	5 meters
410	10 meters
420	20 meters
450	50 meters
475	75 meters
499	99 meters
4XX	Custom Length XX meters

Cable construction

The following illustration shows the pin-out of theTime Link cable.

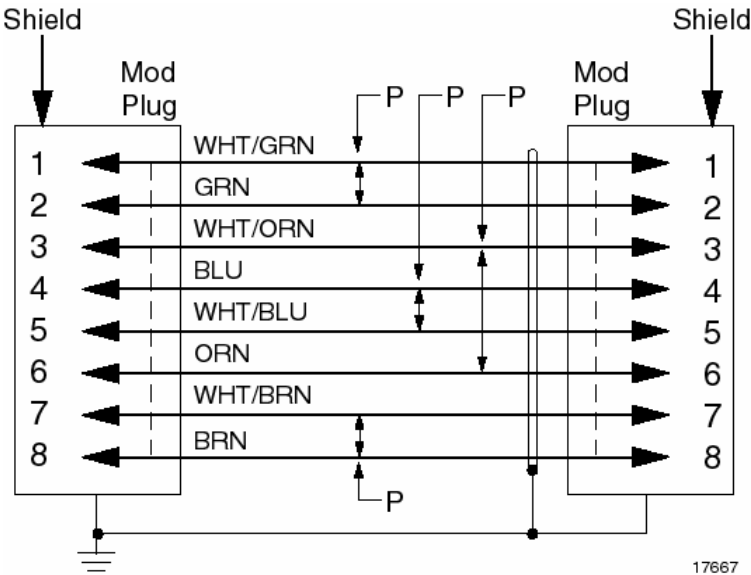


Figure 3: Cable Construction and Pin Assignments

Other Time Link Components

The installation of the cable components requires these additional T-connectors and modular terminators.

Table 2: Time Link T-Connector and Modular Terminator

QTY	Description	Part Number
2	RJ45 (8x8) Modular Plug Terminator (100 Ohm)	51192707-100
AR	T Connector - Receptacle RJ45 - 2x8P8C to 8P8C Shielded	51192708-100

1. GPS / Time Link Components

2.0. Time Link Cables

3. Time Link Cable Diagnostics

3.7 Time Link Cable Diagnostic Aide

Time Link Signals

A diagnostic tool is available that provides a quick visual indication of the current conditions of the 4 signals (2 pairs of differential and 2 non-differential signals) listed in the table below:

Table 3: Time Link Cable Signals

Signal	Description
1PPS (+/-)	Differential Pair – Provides synchronization reference for TPS Master clock (GUS / APP) and sync reference for TPN and Segment Masters in TPN Bridge application.
TSIP (+/-)	Differential Pair – Provides the current time in Trimble TSIP format when connected to a GPS antenna. Provides current time in Time Link Packet format when generated by a TPN Master (no GPS_DETECT active).
TMACT	Non-differential – This level (low true) indicates that a TPN Bridge Time Link Master (TPN Master) is active.
GPS_DETECT	Non-differential – This level (low true) indicates that a cable connection to the GPS power junction box is present. The GPS antenna is used as a timing reference.

Using the diagnostic aide

The operation of the Time Link cable is dependent upon the proper connection of all of the signals from the source to the receivers within the modular RJ45 components that combine to create the Time Link cable system.

The Time Link Cable Diagnostic Aide may be useful during installation of the Time Link cables, T-connectors and Terminators to provide a quick visual indication of complete signal path connectivity. Since the cable is routed in a daisy chain manner, a natural place to perform the first signal checks is at the far end(s) of the time link cable. The Time Link Diagnostic Aide contains built in termination, so it may be used in place of

2. Time Link Cable Diagnostics

1.0. Time Link Cable Diagnostic Aide

the modular terminators or in addition to the modular terminators at any Modular RJ45 jack along the length of the cable.

The Time Link Cable Diagnostic Aide may be used in GPS applications for GUS and APP nodes, in TPN Bridge applications with Time Link and no GPS and in TPN Bridge applications with GPS / Time Link.

To check for the proper signal indications at any point along the cable system, simply unplug a Time Link cable or terminator and insert the RJ45 plug of the Diagnostic Aide. Monitor the LED activity in accordance with Table 4 and Table 5.

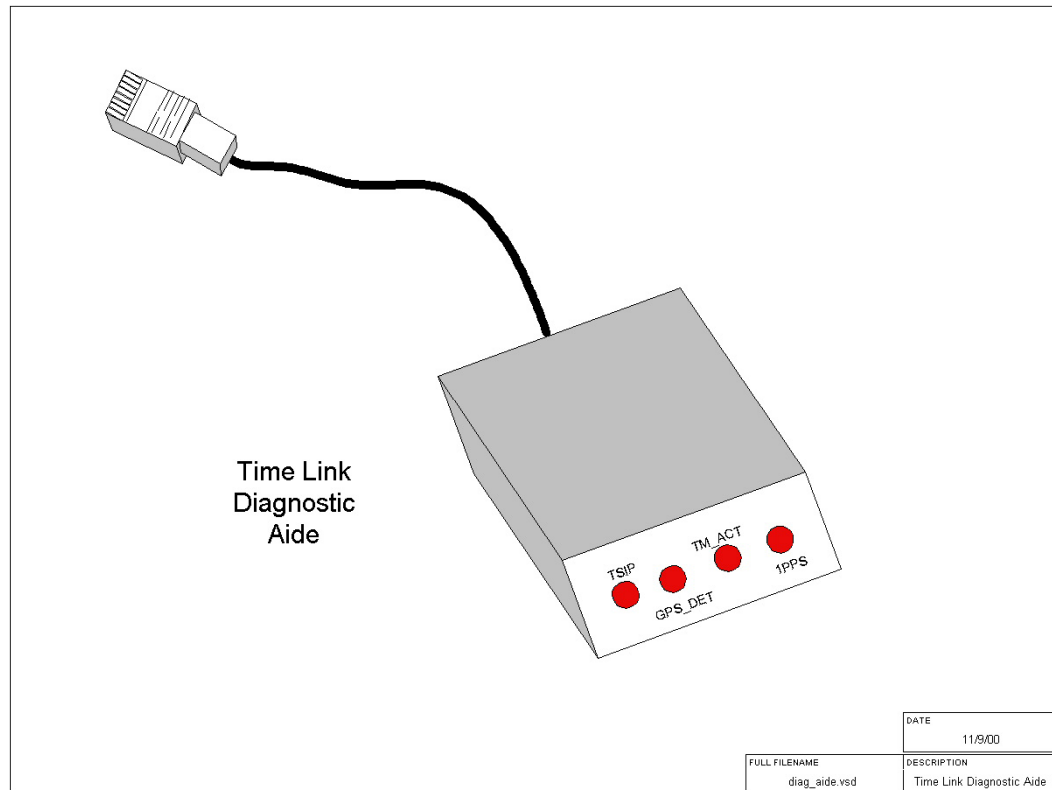


Figure 4 GPS/Time Link Diagnostic Aide

Diagnostic indicators

The following tables show the expected operation for the LEDs in the various configurations.

When connected to an active GPS source the Indicators as shown in Table 4 should be displayed. This applies to TPN Bridge or GUS/APP nodes.

Table 4: LEDs with Active GPS Signals

LEDs with Active GPS Signals (GUS / APP / TPN Bridge)	
LED	Operation
1PPS	Flashing at a rate of 1 Flash per second.
TSIP	Flashing at a rate of 1 Flash per second.
TM_ACT	OFF
GPS_DET	Continuously ON

When connected to an active TPN Master, the following Indicators should be displayed. This is unique to TPN Bridge only.

Table 5: LEDs with Active TPN Master & No GPS (TPN Bridge)

LEDs with Active TPN Master & No GPS	
LED	Operation
1PPS	Flashing at a rate of 1 Flash per second.
TSIP	Flashing at a rate of 1 Flash per second.
TM_ACT	Continuously ON
GPS_DET	OFF

2. Time Link Cable Diagnostics

1.0. Time Link Cable Diagnostic Aide

3. Troubleshooting Tips

3.7 CLOKSTAT Schematic

Description

The “CLOKSTAT” schematic, as shown in Figure 5, is used extensively to determine the overall Synchronization state of the TPS nodes. Each node is listed in sequence of lowest to highest node address along with its Mode of operation (LOCAL, LISTENER, SLAVE and MASTER), cable selection status (normally ALTCLBLA or ALTCLBLB), synchronization state (PRECREF, DIGSYNCH, SUBSYNCH, NOSYNC) and translator status (NONTR, TRNSLTR).

Node Address	CLK_MOD	CLK_CBL	CLK_SYN	CLK_TRN
ND001				
ND002	LISTENER	ALTCLBLB	DIGSYNCH	NONTR
ND003				
ND004	MASTER	ALTCLBLA	PRECREF	NONTR
ND005	LISTENER	ALTCLBLA	DIGSYNCH	NONTR
ND006				
ND007				
ND008				
ND009				
ND010				
ND011				
ND012				
ND013				
ND014				
ND015				
ND016				
ND017				
ND018				
ND019				
ND020				
ND021				
ND022				
ND023				
ND024				
ND025				
ND026				
ND027				
ND028				
ND029				
ND030				
ND031				
ND032				
ND033				
ND034				
ND035				
ND036				
ND037				
ND038				
ND039				
ND040				
ND041				
ND042				
ND043	LISTENER	ALTCLBLA	DIGSYNCH	NONTR
ND044				
ND045				
ND046				
ND047				
ND048				
ND049				
ND050				
ND051	SLAVE	ALTCLBLB	DIGSYNCH	NONTR
ND052				
ND053				
ND054				
ND055				
ND056				
ND057				
ND058				
ND059				
ND060				
ND061				
ND062				
ND063				
ND064				
ND065				
ND066				
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ND081				
ND082				
ND083				
ND084				
ND085				
ND086				
ND087				
ND088				
ND089				
ND090				
ND091				
ND092				
ND093				
ND094				
ND095				
ND096				

Figure 5: CLOKSTAT Schematic

3. Troubleshooting Tips

1.0. CLOKSTAT Schematic

Interpreting the schematic

The training module publication, *Interpret System Clock Functions* (L61524 LCN), provides a very detailed explanation of the fields listed above and that level of detail is not repeated here. The most important column is left most position. The user should look for a single node as MASTER, a single node as SLAVE and all other nodes as LISTENERS *except* in specific TPN Bridge applications where the TPN Bridge nodes have been configured as both TPN and Segment Masters. In this case, the NCF assignments will show the node addresses of the Bridge Nodes as clock sources. Since the bridge nodes do not participate *directly* in the reporting associated with the updates of the CLOKSTAT schematic, these nodes do not show up as MASTER and SLAVE. However, when the system is operating in the expected manner, all other nodes should still report as LISTENER.

2.0 TDC Status Display

Introduction

The application that generates this display is installed on a TPN Bridge node. This application is called the TPN Bridge Viewer. When invoked, you choose the “View Bridge Status” button to get the “TDC Status” display as shown in Figure 6.

TDC Status

System Status

☐ Loopback Mode
☐ LAT Update Ready
☐ Recompute LAT
☐ Interrupt NT
☐ Integrity Check
☐ Idle Message
☐ Mixed System
☐ Recording
☐ LCNI Net Test
☐ Time Testing
☐ Silence Non-Redun
☐ Silence Redun
☒ Physical Dirty
☐ Calculate Rates
☒ Looping
☐ Multi-Seg Time Test
☐ Timer - 1 SEC
☐ Timer - 50MS
☐ Disconnected
☐ LCNI Media Dead
☐ Alone In Ring
☐ Awaiting Secondary
☒ Reserved Bit 22
☐ Reserved Bit 23
☐ Reserved Bit 24
☐ Reserved Bit 25
☐ Reserved Bit 26
☐ Reserved Bit 27
☐ Reserved Bit 28
☐ Reserved Bit 29
☐ Reserved Bit 30
☐ Reserved Bit 31

Value: 000A0200

Update Enabled at
1 milliSecond

Refresh Rate Done

Notice Status

☐ Stack Warning
☐ Safe Area Corruption
☐ LCNI Word 2 Error
☐ LCNI Word A Error
☐ LCNI Out IOCB Error
☐ LCNI In Chan 1 Error
☐ LCNI In Chan 2 Error
☐ Tx IOCB Not Done
☐ Clock Cmd Buf Ovrflw
☐ Multi Alone In Ring
☐ Reserved Bit 10
☐ Reserved Bit 11
☐ Reserved Bit 12
☐ Reserved Bit 13
☐ Reserved Bit 14
☐ Reserved Bit 15

Value: 0000

Cable Status

☐ In Use: 0=A, 1=B
☒ Cable Master
☒ Future Master
☒ Send Crossed
☐ Reserved Bit 4
☐ Reserved Bit 5
☐ Reserved Bit 6
☐ Reserved Bit 7
☐ Reserved Bit 8
☐ Reserved Bit 9
☐ Reserved Bit 10
☐ Reserved Bit 11
☐ Reserved Bit 12
☐ Reserved Bit 13
☐ Reserved Bit 14
☐ Reserved Bit 15

Value: 6000

Output Status

☐ Need Alive
☐ Need Resync
☐ Need Ping
☐ Need Redund
☐ Need Stun
☐ Need Cable Swap
☐ Need Ring Init
☐ Need Kill
☐ Reserved Bit 8
☐ Reserved Bit 9
☐ Reserved Bit 10
☐ Reserved Bit 11
☐ Reserved Bit 12
☐ Reserved Bit 13
☐ Reserved Bit 14
☐ Have Need

Value: 0000

Redund Status

☒ Config Primary
☒ Running Primary
☒ Hearing LCNI
☒ Hearing NT
☒ Heard LCNI
☐ LCNI Receiver
☐ Failover Testing
☐ Failover Stall
☐ Reserved Bit 8
☐ Reserved Bit 9
☐ Reserved Bit 10
☐ Reserved Bit 11
☐ Reserved Bit 12
☐ Reserved Bit 13
☐ Reserved Bit 14
☐ Reserved Bit 15

Value: F800

Time Status

☐ Analyze
☐ Sync Started
☐ Sync Ended
☐ Master Enable
☒ Init Done
☐ Clock Unbusy
☒ Am Synched
☐ Ignore Synch
☒ Bridge Mode
☒ Enhanced Mode
☐ Reserved Bit 10
☐ TL Port Disabled
☐ Sync Queued
☐ Reserved Bit 13
☐ Reserved Bit 14
☐ Reserved Bit 15

Value: 1AC0

Time Mastership

☒ Segment Master
☒ TPN Master
☐ Reserved Bit 2
☐ Reserved Bit 3
☐ Reserved Bit 4
☐ Reserved Bit 5
☐ Reserved Bit 6
☐ Reserved Bit 7
☐ Reserved Bit 8
☐ Reserved Bit 9
☐ Reserved Bit 10
☐ Reserved Bit 11
☐ Reserved Bit 12
☐ Reserved Bit 13
☐ Reserved Bit 14
☐ Reserved Bit 15

Value: C000

Figure 6: TDC Status Display (TPN Bridge)

3. Troubleshooting Tips

2.0. TDC Status Display

Status display description

The TDC Status Display (TPN Bridge Nodes Only) contains information regarding the synchronization state of the particular Bridge node on which the display is presented. This display provides a method of determining the health of TPN Bridge time synchronization from the perspective of a single Bridge node. The important fields in this display (regarding Time Synchronization) are presented in the columns identified as “Time Status” and “Time Mastership”.

Table 6 shows the items that are checked in the Time Status entity for normal conditions as shown in TDC Status dialog page (see Figure 6).

Table 6: TPN Bridge Time Status Check Box Status (Normal Conditions)

	TPN and Segment Master Primary	TPN and Segment Master Secondary	TPN Master Primary	TPN Master Secondary	Segment Master Primary / Secondary
Analyze					
Sync Started					
Sync Ended					
Master Enable	X		X		
Init Done	X	X	X	X	X
Clock Unbusy					
Am Synched	X	X	X	X	X
Ignore Sync					
Bridge Mode	X	X	X	X	X
Enhanced Mode	X	X	X	X	X
TL Port Disabled					
Sync Queued					

3.0 TPS Functions for Synchronizing Time across Nodes

Review sequence

Here is a quick review of TPS time synchronization functions, independent of the GPS system. The TPN Bridge uses additional synchronization functions and is discussed later.

6. Each LCN can be configured to have two nodes (call them A and B) that can act as clock sources. The active source is called the "MASTER"; the standby source is called the "SLAVE."
6. The user sets the local time into the MASTER (say A) which sends it to all other nodes, called "LISTENERS". If the node (A) should cease the MASTER function, the SLAVE (B) will assume the role of MASTER.
6. When the failed clock source node (A) returns to service, it will assume the SLAVE role; in other words, the system does not automatically return the MASTER role to the original node. If the current MASTER (B) should fail, the SLAVE (A) will then become MASTER. There is no way to cause the 2 clock source nodes to switch roles other than by the node in the MASTER role failing to perform that function.
6. If both clock source nodes are down, every other node will go to "LOCAL" mode, i.e., use its local clock, whose stability is +/- 50 ppm. This means the time drift between nodes could be up to +/- 1.8 seconds/day. (Likewise, the time drift between separate LCNs that are working properly could be up to +/- 1.8 seconds/day.)
6. Nodes will also go to LOCAL temporarily when the MASTER role is being switched to the other node.
6. No alarms or other notifications are issued by the system on any of these events. However, the clock status of all nodes can be seen on the CLOKSTAT schematic (MASTER, SLAVE, LISTENER, or LOCAL).

3.7 Definition of Terms for Discussing GPS Time Mastership

New terminology

Since GPS adds a new dimension to TPS, some new terminology is needed for the purposes of this document.

- *LCN MASTER/SLAVE*: A clock source node not connected to a GPS system (i.e., uses its local clock).

3. Troubleshooting Tips

5.0. Diagnosing Loss of GPS signal

- **GPS MASTER/SLAVE:** A clock source node connected to a GPS system and receiving UTC.
- **NGPS MASTER/SLAVE:** A clock source node connected to a GPS system but not receiving the UTC signal (e.g., GPS system failed, time link cable broken, etc.). Such a node cannot become either a *GPS* or *LCN* MASTER, because it is forever looking for the absent UTC signal. Either the GPS system must return to service, or the node must be disconnected from GPS (unplug the time link cable from the LCNP4 card).

The CLOKSTAT display shows only the terms "MASTER" and "SLAVE". For a node connected to GPS, the presence or absence of a blinking yellow LED on its LCNP4 card indicates if it is receiving UTC or not.

5.0 Diagnosing Loss of GPS signal

The effect of losing the GPS 1PPS and TSIP packets is somewhat different in the TPN Bridge application from the GUS / APP Node application. The Non-Bridge discussion is presented first.

GUS / APP Node (RNOS Clock Driver Operation)

Lets say that the clock Master is running in a normal manner with a connection to an active 1PPS and TSIP packets from the GPS antenna. Lets further say that the 1PPS / TSIP signals are then lost because of a loss of antenna power or some other reason. Lets also say that the connection to the power junction box is intact. What will happen?

The first visible sign will be the loss of the 1PPS flashing LED.

The clock hardware "left on its own" wants to maintain itself as Master and continue to send Time Frames onto the LCN. It also wants to report the absence of the GPS reference through its status. It is this report of "loss of sync" to the reference that makes the RNOS Clock Driver software take the clock hardware out of *GPSTMaster* mode and results in sending it through a periodic sync-up process.

6.0 GPS System Configurations and Failure/Recovery Characteristics (Non-TPN Bridge)

Basic configurations

There are three basic configurations for the GPS system. The following lists the three basic configurations. They are followed by a detailed description of each.

- A GPS system is connected only to one clock-source node (A) and each of one or more LCNs.
- A GPS system is connected to both clock-source nodes (A) and (B) on each of one or more LCNs.
- Two separate GPS systems are used, one connected to one clock-source node (A) on each of one or more LCNs, and the other connected to the other clock-source node (B) on each of the same LCNs.

Configuration 1

A GPS system is connected only to one clock-source node (A) and each of one or more LCNs.

Characteristics of configuration 1:

- Initially, the node (A) on each LCN connected to the GPS system will be in *GPS MASTER* mode. The other clock-source node (B) will be in *LCN SLAVE* mode.
- If a *GPS MASTER* node (A) should fail, the *LCN SLAVE* (B) on that LCN will become *LCN MASTER*, so that LCN will become locally synchronized (as before GPS was installed). Presumably there would be alarms or other indications that node (A) had failed.

When the failed node (A) returns to service, it will be as *GPS SLAVE*. In order for node (A) to again become *GPS MASTER* and synchronize its LCN to UTC, it will be necessary to remove the *LCN MASTER* node (B) from service briefly to force the switchover.

- If the GPS system should fail, node (A) on each LCN will go to *NGPS SLAVE* mode; the *LCN SLAVE* (B) on each LCN will become *LCN MASTER*, so each LCN will become locally synchronized.

When the GPS system returns to service, its node (A) on each LCN will go to *GPS SLAVE*. In order for node (A) to again become *GPS MASTER* and synchronize its

3. Troubleshooting Tips

6.0. GPS System Configurations and Failure/Recovery Characteristics (Non-TPN Bridge)

LCN to UTC, it will be necessary on each LCN to remove the *LCN MASTER* node (B) from service briefly to force the switchover.

- CLOKSTAT will show whether node (A) or (B) is MASTER or SLAVE. For the node (A) connected to GPS, presence or absence of its blinking yellow LED indicates if it is a *GPS* or *NGPS* MASTER/SLAVE.

The user should monitor CLOKSTAT periodically. If node (B) is shown as "MASTER", then it is in the *LCN MASTER* role, and synchronization to UTC is not occurring. In that case, if the yellow LED of node (A) is not blinking, the GPS system has failed and needs service; if the LED is blinking, the GPS system probably experienced a momentary failure but is OK now, so node (B) should be temporarily removed from service so that node (A) can again take the role of *GPS MASTER*.

- The most suitable nodes for this configuration would be GUS rather than APP, since GUS nodes can more readily be removed from service with minimum impact on the system.

Configuration 2

A GPS system is connected to both clock-source nodes (A) and (B) on each of one or more LCNs.

Characteristics of configuration 2:

- If the *GPS MASTER* node (A) on an LCN should fail, the *GPS SLAVE* (B) will become *GPS MASTER*.

When node (A) returns to service, it will be as *GPS SLAVE*. There is no need to switch the MASTER role back to node (A).

- If the GPS system should fail, all nodes on each LCN will go to LOCAL. (Both nodes (A) and (B) will try to become *GPS MASTER* but fail, so both will then go to LOCAL.) While in this state, the CLOKSTAT schematic will show all nodes in LOCAL, and the yellow LEDs will not flash.

When the GPS system returns to service, those nodes will resume their *GPS MASTER/SLAVE* roles.

- The user should monitor the CLOKSTAT schematic and LEDs periodically to detect these conditions.

If the GPS system is out of service for a long time, it may be desirable to return to local LCN synchronization. This can be done by unplugging the Time Link cable

from one of the clock source nodes (say, B) on each LCN. This will allow that node (B) to become *LCN MASTER*; the other node (A) will become *NGPS SLAVE*. After the GPS system is restored to service, the flashing yellow LED will be seen on node (A), and the Time Link cable may be re-connected to node (B). Both nodes will automatically return to *GPS MASTER* and *SLAVE* roles.

- Either GUS or APP nodes would be suitable for this configuration.

Configuration 3

Two separate GPS systems are used, one connected to one clock-source node (A) on each of one or more LCNs, and the other connected to the other clock-source node (B) on each of the same LCNs.

Characteristics of configuration 2:

- If the *GPS MASTER* node (A) on an LCN should fail, the *GPS SLAVE* (B) will become *GPS MASTER* and will continue to synchronize all nodes on its LCN to UTC using the second GPS system.

When node (A) returns to service, it will be as *GPS SLAVE*. There is no need to switch the *MASTER* role back to node (A).

- If the GPS system connected to the *GPS MASTER* node (A) on each LCN should fail, that node (A) will become *NGPS SLAVE*, while each *GPS SLAVE* node (B) will become *GPS MASTER* and continue to synchronize all nodes on its LCN to UTC using the second GPS system.

When the failed GPS system returns to service, its node (A) on each LCN will become *GPS SLAVE*. There is no need to switch the *MASTER* role back to node (A).

- If the GPS system connected to the *GPS SLAVE* node (B) on each LCN should fail, that node (B) on each LCN will go to *NGPS SLAVE* mode.

When the failed GPS system returns to service, its node (B) on each LCN will return to *GPS SLAVE* mode.

- If both GPS systems should fail, all nodes on each LCN will go to *LOCAL*. (Both nodes (A) and (B) will try to become *GPS MASTER* but fail, so both will then go to *LOCAL*.) While in this state, the CLOKSTAT schematic will show all nodes in *LOCAL*, and the yellow LEDs will not flash.

When either of the GPS systems is returned to service, its node on each LCN will go to *GPS MASTER* mode, while the nodes on the still-failed GPS system will go to

3. Troubleshooting Tips

6.0. GPS System Configurations and Failure/Recovery Characteristics (Non-TPN Bridge)

NGPS SLAVE mode. Each LCN would then again be synchronized to UTC. When the other GPS system returns to service, its nodes on each LCN will go to *GPS* SLAVE mode.

- The user should monitor the CLOKSTAT schematic and LEDs periodically to detect these conditions.
- While either GUS or APP nodes could be used in this configuration, APP nodes might be a better choice since they are less likely to be removed from service.

The result is that the clock Master will be demoted to Local mode. If both clock sources are connected to the GPS antenna by means of the Time Link cable, then both will be demoted to LOCAL Mode until the signal is restored or the Time Link cable is disconnected from one of the two clock source nodes.

If only one clock source is connected to the GPS antenna, then the other clock source will become Master clock without reference to UTC.

The CLOKSTAT schematic will show the result of this activity by means of the Clock Mode and sync status columns.

TPN Bridge GPS Loss of Signal failure/recovery characteristics

In this configuration, the GPS Antenna is attached to the TPN / Segment Master Primary and Secondary Bridge Nodes by means of the Power Junction box and Time Link cable.

The Time Link cable may be connected just to this TPN / Segment Master Bridge pair, or it may be connected to the TPN / Segment Master Bridge pair and additional Bridge pairs configured as Segment Masters.

The TPN Bridge design provides for an alternate means of LCN segment synchronization in the event that the GPS system goes off line. When the Clock driver software in any Bridge node detects that proper signaling is not being received through the Time Link port, a command is given to the clock hardware to disregard the Time Link port and perform synchronization using time messaging via LAN transmissions from the Primary TPN Master. Users will observe this condition in the following manner:

- If the GPS system connected to the TPN Bridge nodes should fail:
 - 4) The Yellow 1PPS LEDs will quit flashing each second.
 - 4) The “Am Synched” check box will go unchecked.
 - 4) The “Time Link Port Disabled” box will be checked.
 - 4) The “Sync Queued” box will be checked.

- 4) The CLOKSTAT schematic will show all nodes momentarily go to LOCAL mode.
- The TPN Bridge nodes will then establish synchronization using the LAN time transmissions for the duration of the re-sync period (4 minutes.) While the system is synchronized using this technique, users will observe the following:
 - 4) No Yellow 1PPS LEDs will be flashing each second.
 - 4) The “Am Synched” check box will be checked.
 - 4) The “Time Link Port Disabled” box will be checked.
 - 4) The “Sync Queued” box will be unchecked.
 - 4) The CLOKSTAT schematic will show all nodes as LISTENER mode.
- When the 4 minute re-sync period has elapsed, the TPN Bridge nodes will attempt to re-establish sync to the GPS reference. Since each TPN Bridge performs this task using an independent timer, this activity is not performed in unison on all TPN Bridge nodes. The Yellow 1PPS LEDs will remain off.
 - 4) The “Am Synched” check-box will go unchecked.
 - 4) The “Time Link Port Disabled” box will go unchecked.
 - 4) The “Sync Queued” box will be checked.
 - 4) The CLOKSTAT schematic will show all nodes shortly go to LOCAL mode.
- If the GPS system is back on line, the sync status check boxes will return to the normal state indicated in Figure 7. If the GPS system is still off line, the user will observe a repeat of the sequence listed above.

7.0 Use of GPS Monitor Port

Overview

Trimble provides a Monitor program for use with the Palisade and Acutime 2000 smart antennas. This program along with a user guide is downloadable from Trimble's Web site. The startup screen and Monitor screens are shown in figures 9 and 10.

To view the Monitor screen data, the user must provide a connection between the Power Junction box Monitor port and an available PC COM port. The signals coming out of the Power Junction box are RS422, so an RS422 to RS232 converter is required.

A number of commercially available converters derive power for the converter directly from the PC COM port through the DTR and RTS connections. The RS232 interface powered converters should list requirements for maximum cable capacitance and minimum assertion time for RTS or DTR. Most of these types of converters also provide a means of supplying external power if the restrictions cannot be met. Please read the power considerations information provided by the supplier of the converter. An example of a common interface is shown in figure 8.

Interface signals

Table 7 shows the RS422 to RS232 signal paths and a common (but not universal) configuration for the RS422 pins in a DB9 connector. Please refer to the supplier data for the supplier's specific connectivity on the RS422 side.

Table 7: Interface Junction Box Monitor Port to PC COM Port

Power Junction Box Monitor Port J3 DB9F		Cable Interface DB9M to DB9M		RS422 to RS232 Converter DB9F to DB9F		PC COM Port DB9M
		<<RS422>>		<<RS232>>		
Signal RS422	DB9 (J3) Pin #	Wire Pairs	DB9	RS422 to RS232 Converter	PC DB9 COM Port	
	---	Pair 1	---		---	
Ground	5		5 (Gnd)	<<>>	5 (Gnd)	
	---		9 (Aux Pwr)	<<>>	9	
Port B	3	Pair 2	8 (Out)	<<<<	3 (TxOut)	
Port B	7		3 (Out)		---	
	---		---	Note 1 <<	7 (+RTS)	
Port B Tx(a)-	2	Pair 3	2 (In)	>>>>	2 (RxIn)	
Port B Tx(b)+	6		7 (In)		---	
	---		---	Note 1 <<	4 (+DTR)	
	Shield	Shield	Shield	Shield	Shield	
Note 1: Converter power is supplied from the PC's RTS or DTR in this example and auxiliary power is often supported at the pin 9 location. Please refer to the supplier data for the supplier's specific connectivity on the RS422 side.						

3. Troubleshooting Tips

7.0. Use of GPS Monitor Port

Web sites

Some web site addresses for converters are provided below:

<http://www.bb-europe.com/>

<http://www.kksystems.com/>

<http://www.connectworld.net/abl/abl5b.html>

<http://www.antona.com/>

Start-up Screen

The start-up screen for the Palisade (Acutime 2000) is shown in Figure 7.

This screen appears at the start of the program. It lets you choose which Palisade port you want to use and which PC com port that the selected Palisade port is connected to. Palisade Monitor is capable of talking to both Palisade ports at the same time, but you must first choose one port to start with.

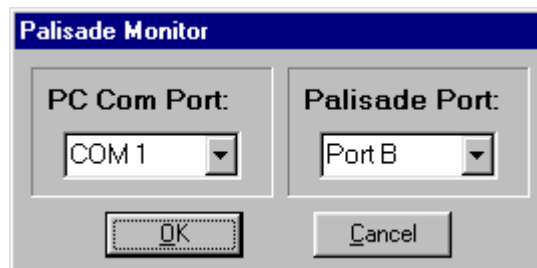


Figure 7 Palisade (Acutime 2000) Monitor Start-up Screen

Monitor display – port B

Shown in Figure 8 is the main screen for Palisade port B. It displays time, position, velocity, SV selection data, and receiver health. The status bar displays Tx and Rx activity, program hints, recording status, firmware version, and com settings. The menu provides other options that let the user send data to the receiver or request additional data from the receiver.

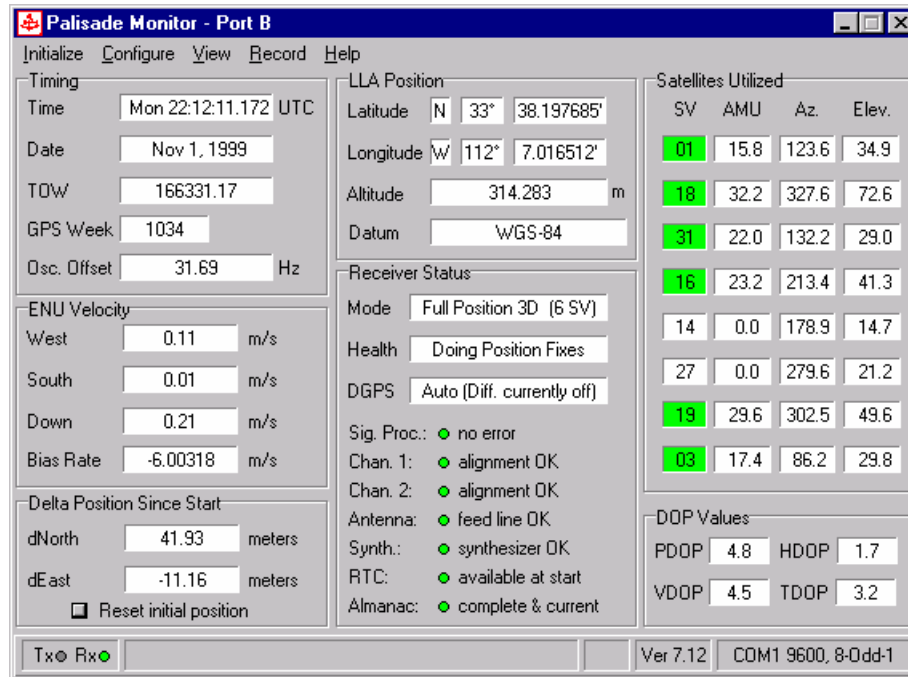


Figure 8 Palisade (Acutime 2000) Monitor Display

Fax Transmittal

Fax No.: (602) 313-4212

Reader Comments

To: Information Development

From: Name

Date:

Title:

Company:

Address:

City:

State:

Zip:

Telephone:

Fax:

Troubleshooting Guide, TP80000

Release **100, 2/03**

Comments:

You may also call 800-343-0228 (available in the 48 contiguous states) or write to:

Honeywell International
Industry Solutions
16404 North Black Canyon Highway
Phoenix, AZ 85053

Time Link/GPS Antenna
Knowledge Building Tools

TotalPlant

Honeywell

Honeywell International
Industry Solutions
16404 North Black Canyon Highway
Phoenix, AZ 85053